



Islamic University of Technology
Lab Report

Experiment No: 04

Student No: 210041105

Course Code: PHY-4242

Name of the Experiment: VERIFICATION OF KIRCHOFF'S
CURRENT LAW (KCL)

Date of Performance: 31/01/23.

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Submitted by:

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Section: 01 (A)

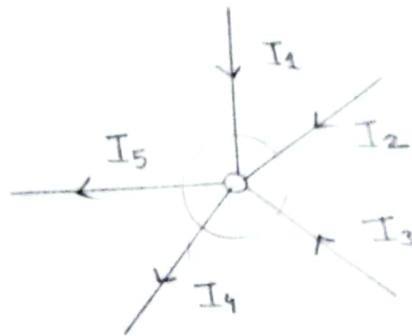
Dept. of Computer Science and Technology (CSE)

Objective:

The objective of this experiment is to verify Kirchhoff's Current Law (KCL) with the help of series-parallel circuit.

Theory:

KCL states that the algebraic sum of currents entering any node equals the sum of the currents leaving the node.

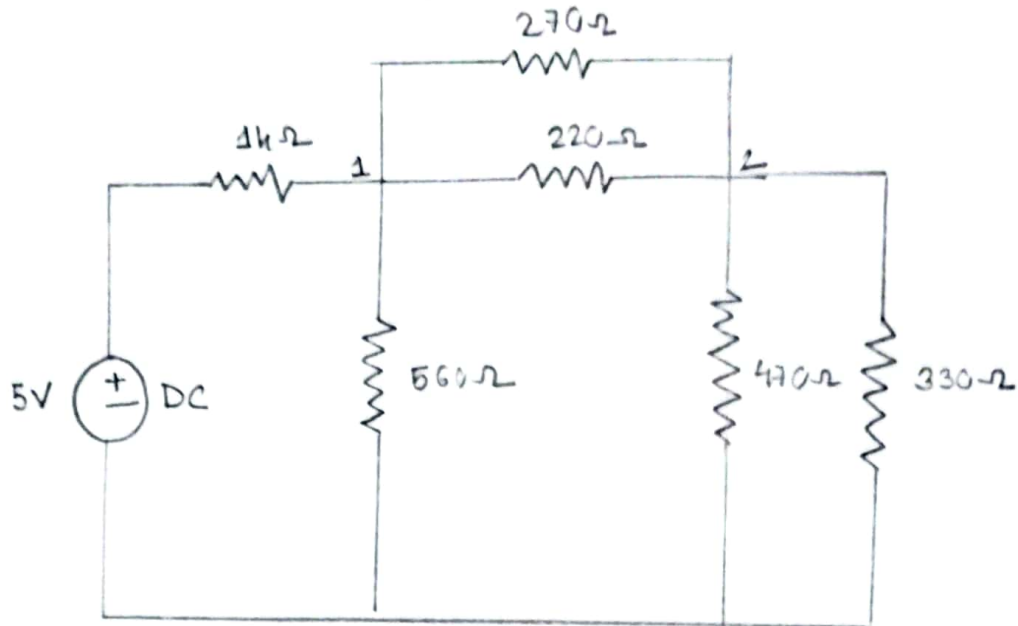


In the figure above,

$$I_1 + I_2 + I_3 = I_4 + I_5$$

$$\Rightarrow I_1 + I_2 + I_3 - I_4 - I_5 = 0$$

Circuit Diagram:



Data Collection:

In practicality, the measured resistances are:

$$R_1 = 992\Omega$$

$$R_2 = 551\Omega$$

$$R_3 = 265\Omega$$

$$R_4 = 216\Omega$$

$$R_5 = 454\Omega$$

$$R_6 = 324.8\Omega$$

And the supplied DC voltage was 5 volts.

Data Calculation:

The current measured through each resistor:

Resistor	Notation	Current
R_1	I_{R_1}	5.1
R_2	I_{R_2}	2
R_3	I_{R_3}	1.4
R_4	I_{R_4}	1.8

Data Analysis:

In node-①, I_{R_1} enters into the node and I_2, I_3 and I_4 comes out from the node.

Following KCL:

$$\begin{aligned} I_{R_1} &= I_{R_2} + I_{R_3} + I_{R_4} \\ &= 2 + 1.4 + 1.8 \\ &= 5.2 \\ &\approx I_{R_1} \end{aligned}$$

Results:

From the above calculation, we have determined that current entering into a node is equal to the current coming out of it.

$$I_1 \approx (I_2 + I_3 + I_4)$$

Discussion:

We have managed to determine the theoretical and practical values of the current flowing through each resistor in the circuit.

We see minor discrepancy in the reading which might have taken place due to instrumental or parallax errors. Kirchhoff's current law is applicable for solving complex electrical circuits. It also introduces the idea of supernodes. Supernodes are formed by enclosing a voltage source connected between two non-reference nodes. While dealing with supernodes, we don't consider the elements connected parallel to it and consider the nodes as a single point for current input and output.